

Monte Carlo Methods

Introduction

Generating Samples

Markov Chain Monte Carlo + Discrepancy

Improving Efficiency

Selected Topics

Git website and repository
Canvas

Fred Hickernell, Fall 2025

Updated 2025 November 13

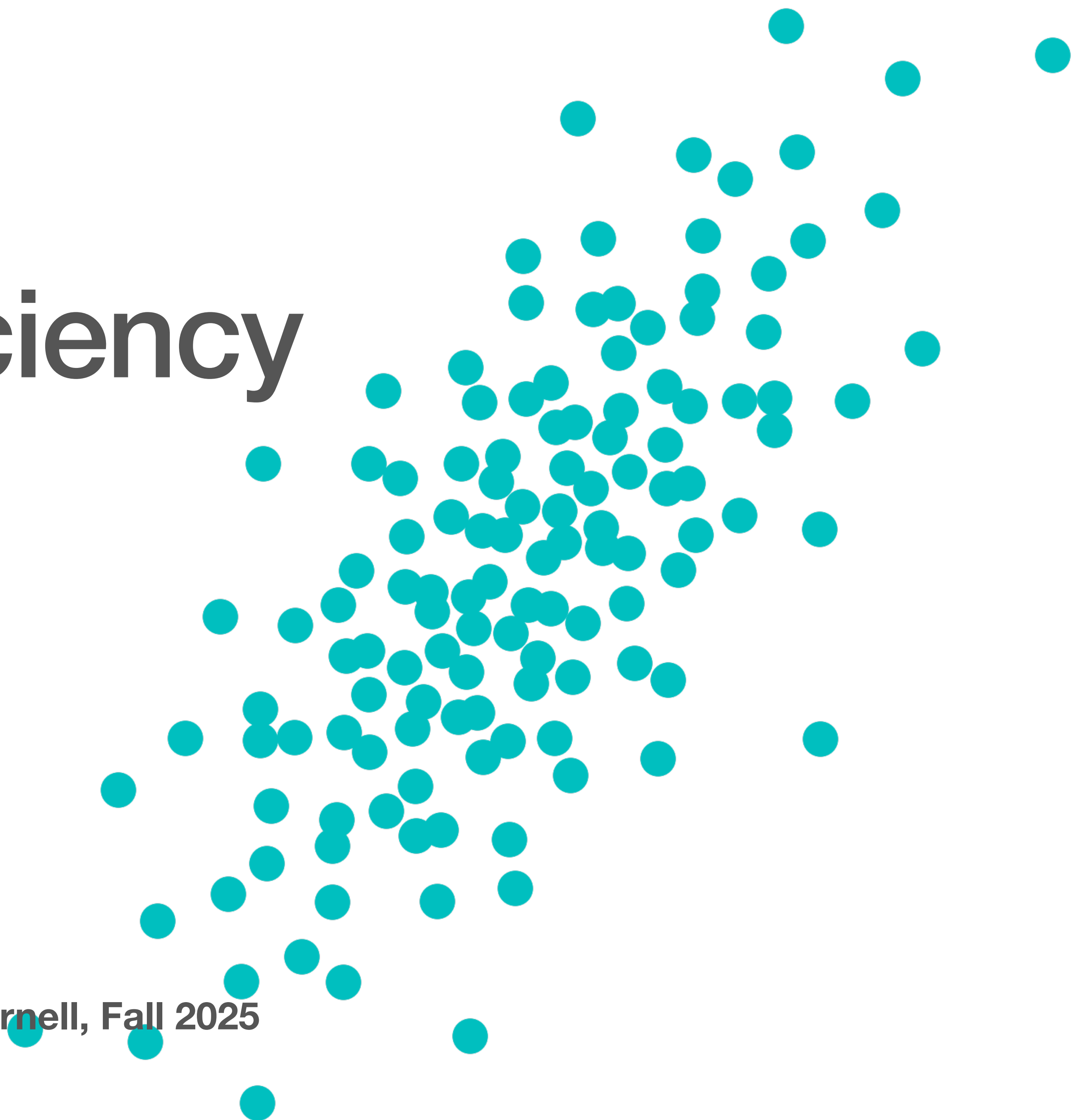


Improving Efficiency

Owen, Chapters 8–10, 15–17

Test 2 Oct 29

MATH 565 Monte Carlo Methods, Fred Hickernell, Fall 2025





Variance Reduction

$$\mu = \mathbb{E}[f(X)] = \mathbb{E}[g(X)], \quad X \sim \mathcal{U}[0,1]^d$$

Is it better to use

$$\hat{\mu}_{f,n} = \frac{1}{n} \sum_{i=0}^{n-1} f(X_i) \quad \text{or} \quad \hat{\mu}_{g,n} = \frac{1}{n} \sum_{i=0}^{n-1} g(X_i), \quad X_0, X_1, \dots \stackrel{\text{iid}}{\sim} \mathcal{U}[0,1]^d$$

to approximate μ ? Since both sample means are unbiased, we can compare their variances (mean squared errors):

$$\text{var}(\hat{\mu}_{f,n}) = \frac{\text{var}(f(X))}{n} \quad \text{var}(\hat{\mu}_{g,n}) = \frac{\text{var}(g(X))}{n}$$

so it depends on the variances of the two functions



Keister example

$$\mu = \int_{\mathbb{R}^d} \cos(\|t\|) \exp(-\|t\|^2) dt$$

$$\mathbf{x} = (\Phi(at_1), \dots, \Phi(at_d)), \quad \Phi \text{ is the CDF of the standard normal}$$

$$d\mathbf{x} = a^d \phi(at_1) \cdots \phi(at_d) dt = \frac{a^d \exp(-a^2 \|t\|^2/2)}{(2\pi)^{d/2}} dt$$

$$t = (\Phi^{-1}(x_1), \dots, \Phi^{-1}(x_d)) / a$$

$$\begin{aligned} \mu = \int_{[0,1]^d} & \frac{(2\pi)^{d/2}}{a^d} \cos\left(\left\|(\Phi^{-1}(x_1), \dots, \Phi^{-1}(x_d))\right\| / a\right) \\ & \times \exp\left(-\frac{2-a^2}{2a^2} \left\|(\Phi^{-1}(x_1), \dots, \Phi^{-1}(x_d))\right\|^2\right) d\mathbf{x}, \quad (a^2 \leq 2) \end{aligned}$$

Whole family $f(\mathbf{x}; a)$ for which $\mu = \mathbb{E}[f(X; a)]$ for $X \sim \mathcal{U}[0,1]^d$



Variable transformations

Suppose that you want to approximate

$$\mu = \int_{\mathcal{T}} g(\mathbf{t}) \, d\mathbf{t}$$

by Monte Carlo. You need to write it as an expectation. Let $\mathbf{t} = \boldsymbol{\psi}(\mathbf{x})$, for some **invertible transformation**, $\boldsymbol{\psi} : \mathcal{X} \rightarrow \mathcal{T}$. Define the **Jacobian determinant** as

$$J(\mathbf{x}; \boldsymbol{\psi}) := \det \left(\left(\frac{\partial \psi_i}{\partial x_j} \right)_{i,j=1}^d \right), \quad J(\mathbf{t}; \boldsymbol{\psi}^{-1}) := \det \left(\left(\frac{\partial \psi_i^{-1}}{\partial t_j} \right)_{i,j=1}^d \right), \quad J(\boldsymbol{\psi}(\mathbf{x}); \boldsymbol{\psi}^{-1}) = \frac{1}{J(\mathbf{x}; \boldsymbol{\psi})}$$

Then

$$\mu = \int_{\mathcal{X}} g(\boldsymbol{\psi}(\mathbf{x})) |J(\mathbf{x}; \boldsymbol{\psi})| \, d\mathbf{x} = \int_{\mathcal{X}} \underbrace{\frac{g(\boldsymbol{\psi}(\mathbf{x}))}{\varrho_{\mathbf{X}}(\mathbf{x})} |J(\mathbf{x}; \boldsymbol{\psi})|}_{f(\mathbf{x})} \varrho_{\mathbf{X}}(\mathbf{x}) \, d\mathbf{x} = \mathbb{E}_{\mathbf{X} \sim \varrho_{\mathbf{X}}} [f(\mathbf{X})]$$



Variable transformations

$$\mu = \int_{\mathcal{T}} g(t) dt = ?$$

Let $t = \psi(x)$, for some **invertible transformation**, $\psi : \mathcal{X} \rightarrow \mathcal{T}$. Then

$$\mu = \int_{\mathcal{X}} g(\psi(x)) |J(x; \psi)| dx = \int_{\mathcal{X}} \underbrace{\frac{g(\psi(x))}{\varrho(x)} |J(x; \psi)|}_{f(x)} \varrho(x) dx = \mathbb{E}_{\mathbf{X} \sim \varrho}[f(\mathbf{X})]$$

If $\mathbf{X} \sim \mathcal{U}[0,1]^d$ (because we know how generate **uniform** samples), then

$$\mu = \int_{[0,1]^d} \underbrace{g(\psi(x)) |J(x; \psi)|}_{f(x)} dx = \mathbb{E}_{\mathbf{X} \sim \mathcal{U}[0,1]^d}[f(\mathbf{X})]$$



Transformations of random variables

Let

- T be a random variable with sample space $\mathcal{T}$ and density ϱ_T
- X be a random variable with sample space $\mathcal{X}$ and density ϱ_X
- $T = \psi(X)$, for some **invertible transformation**, $\psi : \mathcal{X} \rightarrow \mathcal{T}$. Then

$$\varrho_T(t) = \varrho_X(\psi^{-1}(t)) |J(t; \psi^{-1})|, \quad \varrho_T(\psi(x)) = \varrho_X(x) |J(\psi(x)); \psi^{-1})| = \frac{\varrho_X(x)}{|J(x; \psi)|}$$

$$\mu = \int_{\mathcal{T}} \underbrace{h(t) \varrho_T(t)}_{g(t)} dt = \mathbb{E}_{T \sim \varrho_T}[h(T)] = \mathbb{E}_{X \sim \varrho_X}[h(\psi(X))] = \int_{\mathcal{X}} h(\psi(x)) \varrho_X(x) dx$$

$$\text{or } = \int_{\mathcal{X}} \frac{h(\psi(x)) \varrho_T(\psi(x))}{\varrho_X(x)} |J(x; \psi)| \varrho_X(x) dx = \int_{\mathcal{X}} h(\psi(x)) \varrho_X(x) dx$$



Importance sampling

Suppose that you want to approximate

$$\mu = \mathbb{E}_{\mathbf{T} \sim \varrho_{\mathbf{T}}} [h(\mathbf{T})] = \int_{\mathcal{T}} h(\mathbf{t}) \varrho_{\mathbf{T}}(\mathbf{t}) \, d\mathbf{t} = \int_{\mathcal{T}} \underbrace{h(\mathbf{t}) \frac{\varrho_{\mathbf{T}}(\mathbf{t})}{\varrho_{\mathbf{X}}(\mathbf{t})}}_{f(\mathbf{t})} \overbrace{\varrho_{\mathbf{X}}(\mathbf{t})}^{\text{likelihood ratio}} \, d\mathbf{t} = \mathbb{E}_{\mathbf{X} \sim \varrho_{\mathbf{X}}} [f(\mathbf{X})]$$

We try to choose $\varrho_{\mathbf{X}}$ to make $\text{var}[f(\mathbf{X})]$ small when doing IID sampling. If $\mathbf{T} = \boldsymbol{\psi}(\mathbf{X})$ for $\mathbf{X} \sim \mathcal{X}[0,1]^d$ —since often find generate non-uniform samples from uniform ones—then

$$\mu = \int_{\mathcal{T}} h(\mathbf{t}) \varrho_{\mathbf{T}}(\mathbf{t}) \, d\mathbf{t} = \mathbb{E}_{\mathbf{T} \sim \varrho_{\mathbf{T}}} [h(\mathbf{T})] = \mathbb{E}_{\mathbf{X} \sim \mathcal{U}[0,1]^d} [h(\boldsymbol{\psi}(\mathbf{X}))] = \int_{[0,1]^d} h(\boldsymbol{\psi}(\mathbf{x})) \, d\mathbf{x}$$



Importance sampling cont'd

Suppose that you want to approximate

$$\mu = \mathbb{E}_{\mathbf{T} \sim \varrho_{\mathbf{T}}} [h(\mathbf{T})] = \int_{\mathcal{T}} h(\mathbf{t}) \varrho_{\mathbf{T}}(\mathbf{t}) \, d\mathbf{t} = \int_{\mathcal{T}} \underbrace{h(\mathbf{t}) \frac{\varrho_{\mathbf{T}}(\mathbf{t})}{\varrho_{\mathbf{X}}(\mathbf{t})}}_{f(\mathbf{t})} \varrho_{\mathbf{X}}(\mathbf{t}) \, d\mathbf{t} = \mathbb{E}_{\mathbf{X} \sim \varrho_{\mathbf{X}}} [f(\mathbf{X})]$$

$$\hat{\mu}_{\text{IS},n} = \frac{1}{n} \sum_{i=0}^{n-1} h(X_i) \frac{\varrho_{\mathbf{T}}(X_i)}{\varrho_{\mathbf{X}}(X_i)}, \quad X_0, X_1, \dots \stackrel{\text{iid}}{\sim} \varrho_{\mathbf{X}}$$
$$\text{var}(\hat{\mu}_{\text{IS},n}) \approx \frac{1}{n(n-1)} \sum_{i=0}^{n-1} \left[h(X_i) \frac{\varrho_{\mathbf{T}}(X_i)}{\varrho_{\mathbf{X}}(X_i)} - \hat{\mu}_{\text{IS},n} \right]^2$$



Ex. Brownian motion with drift

In computational finance certain outcomes, such as positive option payoffs, may be **rare**

- Adding a drift to the Brownian motion **increases** the probability that asset paths yield positive payoffs and
- **Reduces the variance** of our estimator of the option price (population mean)
- Paths with positive payoffs receive **smaller sample weights** to compensate and keep the mean the same

$$\text{price} = \int_{\mathbb{R}^d} \text{payoff}(\mathbf{x}) \varrho(\mathbf{x}) d\mathbf{x}, \quad \varrho(\mathbf{x}) = \frac{\exp(-\mathbf{x}^\top \Sigma^{-1} \mathbf{x} / 2)}{\sqrt{(2\pi)^d |\Sigma|}}, \quad \Sigma = \left(\min(t_i, t_j) \right)_{i,j=1}^d$$

Here the density ϱ is a discretized **Brownian motion**



Ex. Brownian motion with drift cont'd

$$\text{price} = \int_{\mathbb{R}^d} \text{payoff}(\mathbf{x}) \varrho(\mathbf{x}) d\mathbf{x}$$
$$\varrho(\mathbf{x}) = \frac{\exp(-\mathbf{x}^\top \Sigma^{-1} \mathbf{x} / 2)}{\sqrt{(2\pi)^d |\Sigma|}}, \quad \Sigma = (\min(t_i, t_j))_{i,j=1}^d$$

We importance sample:

$$\varrho_{\text{drift}}(\mathbf{x}) = \frac{\exp(-(\mathbf{x} - \mathbf{a})^\top \Sigma^{-1} (\mathbf{x} - \mathbf{a}) / 2)}{\sqrt{(2\pi)^d |\Sigma|}}, \quad \mathbf{a} = \alpha(t_1, \dots, t_d)^\top$$

$$\text{likelihood ratio} = \frac{\varrho(\mathbf{x})}{\varrho_{\text{drift}}(\mathbf{x})} = \exp(-\alpha x_d + \alpha^2 t_d / 2)$$

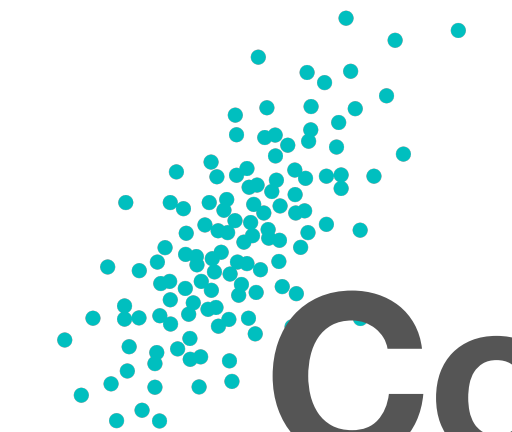
$$\text{price} = \int_{\mathbb{R}^d} \text{payoff}(\mathbf{x}) \exp(-\alpha x_d + \alpha^2 t_d / 2) \varrho_{\text{drift}}(\mathbf{x}) d\mathbf{x}$$

Note that the weight gets **smaller** as αx_d increases or $\alpha^2 t_d$ decreases



Variable transformation / importance sampling observations

- Variable transformation and importance sampling are **intertwined**
- The **choice** of sampling density or variable transformation affects the variance of the sample mean
- You must choose the variable transformation so that
 - The Jacobian determinant, $J(\mathbf{x}; \boldsymbol{\psi})$, is **finite**
 - The probability density, $q_{\mathbf{X}}(\mathbf{x})$, does not **vanish** unless integrand vanishes
- You might try **pilot** samples to help choose transformation



Control variates

$$\mu = \mathbb{E}[f(X)] \iff \mu = \mathbb{E}[f(X) - \beta_1 \eta_1(X) - \dots - \beta_m \eta_m(X)]$$

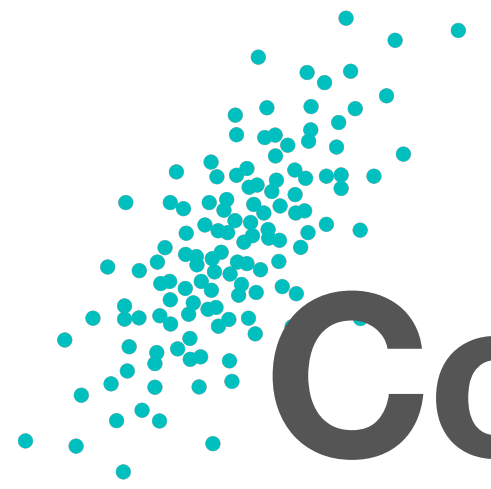
provided that $\mathbb{E}[\eta_1(X)] = \dots \mathbb{E}[\eta_m(X)] = 0$. Want to arrange such that

$$\text{var}[f(X) - \beta_1 \eta_1(X) - \dots - \beta_m \eta_m(X)] \leq \text{var}[f(X)]$$

Then

$$\hat{\mu}_{\text{CV},n} = \frac{1}{n} \sum_{i=0}^{n-1} [f(X_i) - \beta_1 \eta_1(X_i) - \dots - \beta_m \eta_m(X_i)] \text{ converges faster than } \hat{\mu}_n = \frac{1}{n} \sum_{i=0}^{n-1} f(X_i)$$
$$\text{var}(\hat{\mu}_{\text{CV},n}) \approx \frac{1}{n(n-1)} \sum_{i=0}^{n-1} [f(X_i) - \beta_1 \eta_1(X_i) - \dots - \beta_m \eta_m(X_i) - \hat{\mu}_{\text{CV},n}]^2$$

How do we **minimize** $\text{var}[f(X) - \beta_1 \eta_1(X) - \dots - \beta_m \eta_m(X)]$?



Control variates (cont'd)

$$\hat{\mu}_{\text{CV},n} = \frac{1}{n} \sum_{i=0}^{n-1} [f(X_i) - \beta_1 \eta_1(X_i) - \cdots - \beta_m \eta_m(X_i)]$$

How do we **minimize** $\text{var}[f(X) - \beta_1 \eta_1(X) - \cdots - \beta_m \eta_m(X)]$? Use least squares regression with centered response and explanatory variables to choose $\hat{\beta}$:

$$\hat{\beta} = \underset{\mathbf{b}}{\text{argmin}} \|\mathbf{y} - \mathbf{H}\mathbf{b}\|^2$$

$$\text{where } \mathbf{y} = \begin{pmatrix} f(\mathbf{x}_0) - \hat{\mu}_n \\ \vdots \\ f(\mathbf{x}_{n-1}) - \hat{\mu}_n \end{pmatrix}, \quad \mathbf{H} = \begin{pmatrix} \eta_1(\mathbf{x}_0) - \hat{\mu}_{\eta_1,n} & \cdots & \eta_m(\mathbf{x}_0) - \hat{\mu}_{\eta_m,n} \\ \vdots & & \vdots \\ \eta_1(\mathbf{x}_{n-1}) - \hat{\mu}_{\eta_1,n} & \cdots & \eta_m(\mathbf{x}_{n-1}) - \hat{\mu}_{\eta_m,n} \end{pmatrix}$$

$$\hat{\mu}_n = \frac{1}{n} \sum_{i=0}^{n-1} f(\mathbf{x}_i), \quad \hat{\mu}_{\eta_1,n} = \frac{1}{n} \sum_{i=0}^{n-1} \eta_1(\mathbf{x}_i), \quad \dots, \quad \hat{\mu}_{\eta_m,n} = \frac{1}{n} \sum_{i=0}^{n-1} \eta_m(\mathbf{x}_i)$$



Least Squares

How do we solve

$$\hat{\beta} = \underset{\mathbf{b}}{\operatorname{argmin}} \|\mathbf{y} - \mathbf{H}\mathbf{b}\|^2 = \underset{\mathbf{b}}{\operatorname{argmin}} (\mathbf{y} - \mathbf{H}\mathbf{b})^T (\mathbf{y} - \mathbf{H}\mathbf{b})$$

Let

$$\mathbf{b} = (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \mathbf{y} + \mathbf{c}$$

Then

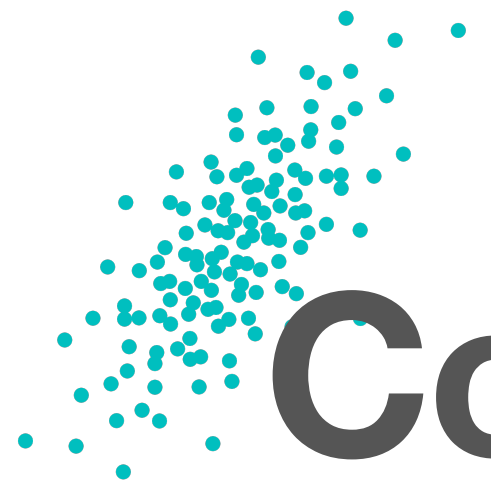
$$\begin{aligned} \|\mathbf{y} - \mathbf{H}\mathbf{b}\|^2 &= \|\mathbf{y} - \mathbf{H}[(\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \mathbf{y} + \mathbf{c}]\|^2 \\ &= \|\mathbf{[I - H(H^T H)^{-1} H^T]y + Hc}\|^2 \\ &= (\mathbf{[I - H(H^T H)^{-1} H^T]y + Hc})^T (\mathbf{[I - H(H^T H)^{-1} H^T]y + Hc}) \\ &= \mathbf{y}^T \mathbf{[I - H(H^T H)^{-1} H^T]y} + \mathbf{c}^T \mathbf{H}^T \mathbf{H} \mathbf{c} \end{aligned}$$

So the optimum comes by setting $\mathbf{c} = \mathbf{0}$



Control variates

- Should be **clever** in choosing the η_j
 - **Good** choices can decrease the sample size required
 - **Bad** choices only hurt in terms of increased **computation** time, but not increased sample size
- The β_j should be chosen by **least squares** for IID sampling
- For **low discrepancy** sequences, the β_j should be chosen to reduce the **variation** (not variance)



Conditional Monte Carlo

Recall that

$$\mathbb{E}(Y) = \mathbb{E}_{\mathbf{Z}}[\mathbb{E}_Y(Y | \mathbf{Z})], \quad \text{var}(Y) = \mathbb{E}_{\mathbf{Z}}[\text{var}(Y | \mathbf{Z})] + \text{var}_{\mathbf{Z}}[\mathbb{E}_Y(Y | \mathbf{Z})]$$

Which implies that

$$\text{var}_{\mathbf{Z}}[\mathbb{E}_Y(Y | \mathbf{Z})] \leq \text{var}(Y), \quad \text{var} \left(\frac{1}{n} \sum_{i=0}^{n-1} \mathbb{E}_Y(Y | \mathbf{Z}_i) \right) \leq \text{var} \left(\frac{1}{n} \sum_{i=0}^{n-1} Y_i \right)$$

Let $(Y = f(\mathbf{X}_{1:d}), \mathbf{X}_{1:d})$ be a vector of random variables

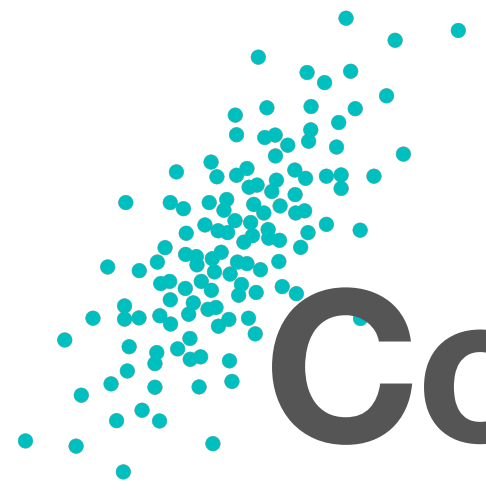
Let $\mathbf{Z} = \mathbf{X}_{2:d}$

If $g(\mathbf{X}_{2:d}) = \mathbb{E}_Y(Y | \mathbf{X}_{2:d})$ has an analytic form, then

$$\mu = \mathbb{E}(Y) = \mathbb{E}_{\mathbf{X}_{2:d}}[g(\mathbf{X}_{2:d})] \text{ and } \text{var}(Y) \geq \text{var}_{\mathbf{X}_{2:d}}[g(\mathbf{X}_{2:d})]$$

If $h(y, \mathbf{X}_{2:d}) = \varrho_{Y|\mathbf{X}_{2:d}}(y | \mathbf{x}_{2:d})$ has an analytic form, then

$$\varrho_Y(y) = \mathbb{E}_{\mathbf{X}_{2:d}}[h(y, \mathbf{X}_{2:d})] \quad \text{probability density expressed as expectation}$$



Conditional Monte Carlo for Density Estimation

Estimate the probability density, ϱ , for $Y = f(X)$, where $X \sim \mathcal{U}[0,1]^d$

If one can identify an **increasing** g , such that

$$y = f(\mathbf{x}) \iff x_1 = g(y; \mathbf{x}_{2:d})$$

Then

$$\begin{aligned} F_{Y|X_{2:d}}(y | \mathbf{x}_{2:d}) &= \mathbb{P}[Y \leq y | X_{2:d} = \mathbf{x}_{2:d}] = \mathbb{P}[f(X) \leq y | X_{2:d} = \mathbf{x}_{2:d}] \\ &= \mathbb{P}[X_1 \leq g(y; \mathbf{x}_{2:d})] = g(y; \mathbf{x}_{2:d}) \end{aligned}$$

$$\varrho(y) = \mathbb{E}[\varrho_{Y|X_{2:d}}(y | \mathbf{x}_{2:d})] = \int_{[0,1]^{d-1}} g'(y; \mathbf{x}) \, d\mathbf{x}$$



Ex. Density of a sum of uniforms

Let $Y = \gamma_1 X_1 + \cdots + \gamma_d X_d$, where $X \sim \mathcal{U}[0,1]^d$ and all the γ_ℓ are non-negative.

What is the **PDF** of Y ?

$$g(y; x_2, \dots, x_d) = \frac{1}{\gamma_1} (y - \gamma_2 x_2 - \cdots - \gamma_d x_d), \quad \gamma_2 x_2 + \cdots + \gamma_d x_d \leq y \leq \gamma_1 + \gamma_2 x_2 + \cdots + \gamma_d x_d$$

$$g'(y; x_2, \dots, x_d) = \frac{1}{\gamma_1}, \quad \gamma_2 x_2 + \cdots + \gamma_d x_d \leq y \leq \gamma_1 + \gamma_2 x_2 + \cdots + \gamma_d x_d$$



Antithetic Sampling

If you are sampling from a probability distribution with symmetry, then you can evaluate f at both X_i and $T(X_i)$, where $T(X)$ has the same distribution as X , e.g.,

$$T(X) = \mathbf{1} - X \text{ for } X \sim \mathcal{U}[0,1]^d \text{ or } T(X) = -X \text{ for } X \sim \mathcal{N}(\mathbf{0}, \Sigma)$$

Note that if X_i are IID, then X_i and $T(X_i)$ are not independent, but $(X_i, T(X_i))$ are

When does $\hat{\mu}_{\text{Anti},n} = \frac{1}{2n} \sum_{i=0}^{n-1} [f(X_i) + f(T(X_i))]$ have a smaller variance than

$$\hat{\mu}_{2n} = \frac{1}{2n} \sum_{i=0}^{2n-1} f(X_i) \text{ for IID } X_i?$$



Antithetic Sampling

Note that

$$\text{var}(\hat{\mu}_{\text{Anti},n}) = \frac{1}{4n^2} \sum_{i=0}^{n-1} \text{var}(f(X_i) + f(T(X_i))) = \frac{\text{var}(f(X))}{2n} [1 + \text{corr}(f(X), f(T(X)))]$$

Whereas

$$\text{var}(\hat{\mu}_{2n}) = \frac{\text{var}(f(X))}{2n}$$

So antithetic sampling helps if $\text{corr}(f(X), f(T(X))) < 0$

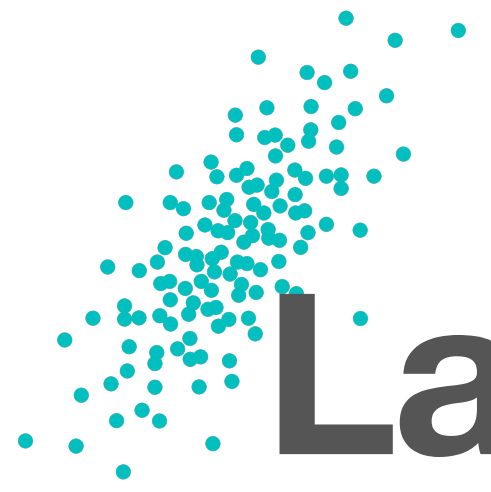
$$\text{var}(\hat{\mu}_{\text{Anti},n}) \approx \frac{1}{4n(n-1)} \sum_{i=0}^{n-1} [f(X_i) + f(T(X_i)) - 2\hat{\mu}_{\text{Anti},n}]^2$$



How to get an antithetic variate

Suppose that you have random variable X with CDF F . One way to construct an **antithetic variate**, $T(X)$:

- Note that $U = F(X) \sim \mathcal{U}[0,1]$
- Also, $1 - U \sim \mathcal{U}[0,1]$
- Therefore $T(X) = F^{-1}(1 - U) = F^{-1}(1 - F(X))$ also has CDF F



Latin Hypercube Sampling in One Dimension

Let's be intentional about placing uniform looking points on $[0,1]$

$$X_i = (i + U_i)/n, \quad U_0, U_1, \dots \stackrel{\text{IID}}{\sim} \mathcal{U}[0,1], \quad i = 0, \dots, n-1, \quad \hat{\mu}_{\text{LHS},n} = \frac{1}{n} \sum_{i=0}^{n-1} f(X_i)$$

Note that $X_i = (i + U_i)/n \sim \mathcal{U}[i/n, (i+1)/n]$, and so

$$\bullet \quad \mathbb{E} [\hat{\mu}_{\text{LHS},n}] = \frac{1}{n} \sum_{i=0}^{n-1} \mathbb{E}[f(X_i)] = \frac{1}{n} \sum_{i=0}^{n-1} \int_{i/n}^{(i+1)/n} f(x) \times n \times dx = \int_0^1 f(x) dx = \mu$$

which means that $\hat{\mu}_{\text{LHS},n}$ is unbiased

- The X_i are not IID but they are independent



Variance of LHS in One Dimension

$$\text{var}(\hat{\mu}_{\text{LHS},n}) = \text{var}\left(\sum_{i=0}^{n-1} \frac{f(X_i)}{n}\right) = \frac{1}{n^2} \sum_{i=0}^{n-1} \text{var}(f(X_i)) \quad \text{since the } X_i \text{ are uncorrelated}$$

$$= \frac{1}{n^2} \sum_{i=0}^{n-1} \int_{i/n}^{(i+1)/n} (f(x) - f(\eta_i))^2 \times n \times dx$$

where $f(\eta_i) = \mathbb{E}[f(X_i)]$ for some η_i by the Mean Value Theorem

$$= \frac{1}{n} \sum_{i=0}^{n-1} \int_{i/n}^{(i+1)/n} (f'(\xi_i(x))(x - \eta_i))^2 dx,$$

where $f(x) = f(\eta_i) + f'(\xi_i(x))(x - \eta_i)$ for some $\xi(x)$ by Taylor's Theorem

$$\leq \frac{1}{n} \sum_{i=0}^{n-1} \frac{\|f'\|_{\infty}^2}{n^3} = \frac{\|f'\|_{\infty}^2}{n^3}$$

$$\text{rmse}(\hat{\mu}_{\text{LHS},n}) \leq \frac{\|f'\|_{\infty}}{n^{3/2}}$$



Latin Hypercube Sampling in d dimensions

$X_{i,\ell} = (\pi_\ell(i) + U_{i,\ell})/n$, $U_{i,\ell} \stackrel{\text{IID}}{\sim} \mathcal{U}[0,1]$, π_ℓ are IID random permutations of $1, \dots, n$

$$\hat{\mu}_{\text{LHS},n} = \frac{1}{n} \sum_{i=0}^{n-1} f(X_{i,1}, \dots, X_{i,d})$$

To quantify uncertainty in the estimate of the mean we need **replications**:

$X_{i,\ell}^{(r)} = (\pi_\ell^{(r)}(i) + U_{i,\ell}^{(r)})/n$, $U_{i,\ell}^{(r)} \stackrel{\text{IID}}{\sim} \mathcal{U}[0,1]$, $\pi_\ell^{(r)} \stackrel{\text{IID}}{\sim}$, $\hat{\mu}_{\text{LHS},n}^{(r)} = \frac{1}{n} \sum_{i=0}^{n-1} f(X_{i,1}^{(r)}, \dots, X_{i,d}^{(r)})$, $r = 1, \dots, R$

$$\hat{\mu}_{\text{LHS},n,R} = \frac{1}{R} \sum_{r=0}^{R-1} \hat{\mu}_{\text{LHS},n}^{(r)} = \frac{1}{nR} \sum_{r,i=0}^{R-1,n-1} f(X_{i,1}^{(r)}, \dots, X_{i,d}^{(r)})$$

$$\text{var}(\hat{\mu}_{\text{LHS},n,R}) \approx \frac{1}{R(R-1)} \sum_{r=0}^{R-1} \left[\hat{\mu}_{\text{LHS},n}^{(r)} - \hat{\mu}_{\text{LHS},n,R} \right]^2$$



Pros and cons of LHS

Pros

- Do not need to **guess** importance sampling distributions or control variates or compute conditional expectations
- Can give **faster** convergence rates than IID

Cons

- Sample size, n , is **fixed**
- Difficult to assess accuracy, except by **replications**; too many replications destroy the faster convergence
- Great convergence rates **do not hold** for multivariate case ($d > 1$)



Low discrepancy sampling

Low discrepancy sampling, also known as [quasi-Monte Carlo methods](#) is an improvement on LHS that can provide [higher order convergence rates](#) than IID sampling for functions with modest smoothness

Recall from earlier the discrepancy, $D(\{\mathbf{x}_i\}_{i=0}^{n-1}, F; K)$, between a point cloud or node set $\{\mathbf{x}_i\}_{i=0}^{n-1}$ and a probability distribution, F , defined by

$$D^2(\{\mathbf{x}_i\}_{i=0}^{n-1}, F; K) = \int_{\mathcal{X} \times \mathcal{X}} K(\mathbf{x}, \mathbf{x}') dF(\mathbf{x}) dF(\mathbf{x}') - \frac{2}{n} \sum_{i=0}^{n-1} \int_{\mathcal{X}} K(\mathbf{x}, \mathbf{x}_i) dF(\mathbf{x}) + \frac{1}{n^2} \sum_{i,j=0}^{n-1} K(\mathbf{x}_i, \mathbf{x}_j),$$

Where we will take the domain to be $\mathcal{X} = [0, 1]^d$



Discrepancy interpretation

- Worst-case error

$$\left| \int_{[0,1]^d} f(\mathbf{x}) \, d(\mathbf{x}) - \frac{1}{n} \sum_{i=0}^{n-1} f(\mathbf{x}_i) \right| \leq D(\mathcal{U}[0,1]^d, \{\mathbf{x}_i\}_{i=0}^{n-1}; K) \|f\|_{\mathcal{H}} \quad \forall f \in \mathcal{H}$$

- Average-case error

$$\mathbb{E}_{f \in \mathcal{GP}(0,K)} \left| \int_{[0,1]^d} f(\mathbf{x}) \, d\mathbf{x} - \frac{1}{n} \sum_{i=0}^{n-1} f(\mathbf{x}_i) \right|^2 = D^2(\mathcal{U}[0,1]^d, \{\mathbf{x}_i\}_{i=0}^{n-1}; K)$$

- **Distance** between the target distribution and the point cloud

$$\begin{aligned} \mathbb{E}_{\mathbf{X}, \mathbf{X}' \sim \mathcal{U}[0,1]^d} K(\mathbf{X}, \mathbf{X}') - 2 \mathbb{E}_{\mathbf{X} \sim \mathcal{U}[0,1]^d, \mathbf{Z} \sim F_{\{\mathbf{x}_i\}_{i=0}^{n-1}}} K(\mathbf{X}, \mathbf{Z}) + \mathbb{E}_{\mathbf{Z}, \mathbf{Z}' \sim F_{\{\mathbf{x}_i\}_{i=0}^{n-1}}} K(\mathbf{Z}, \mathbf{Z}') \\ = D^2(\mathcal{U}[0,1]^d, \{\mathbf{x}_i\}_{i=0}^{n-1}; K) \end{aligned}$$



van der Corput sequence

In one-dimension, evenly spaced points may be constructed as

$$x_i = i/n, \quad i = 0, \dots, n - 1$$

But you need to **fix n** in advance. The **van der Corput sequence** is an infinite sequence that fills the unit interval $[0, 1]$ evenly and one can stop at any n and even keep going. Let b (normally $b = 2$) be a prime base. Then

$$\phi_b(i) = \phi_b(\dots + i_2 b^2 + i_1 b + i_0) = \phi_b((\dots i_2 i_1 i_0)_b) := {}_b(0.i_0 i_1 i_2 \dots) = \frac{i_0}{b} + \frac{i_1}{b^2} + \frac{i_2}{b^3} + \dots$$

The set $\{i/b^m\}_{i=0}^{b^m-1}$ and the set $\{\phi_b(i)\}_{i=0}^{b^m-1}$ have the **same points** but in a different order, but the elements of the second set do not change their order when m increases

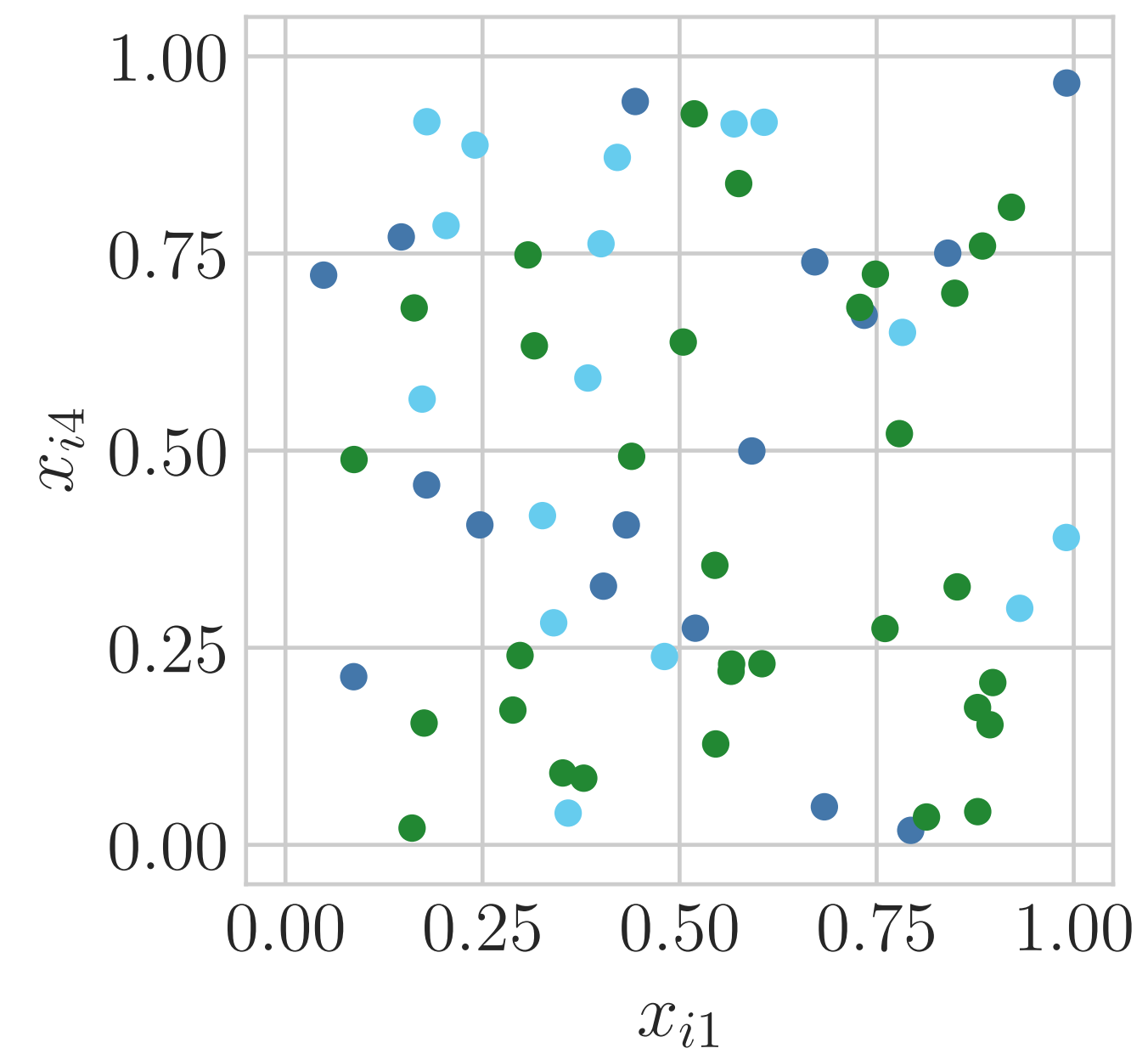
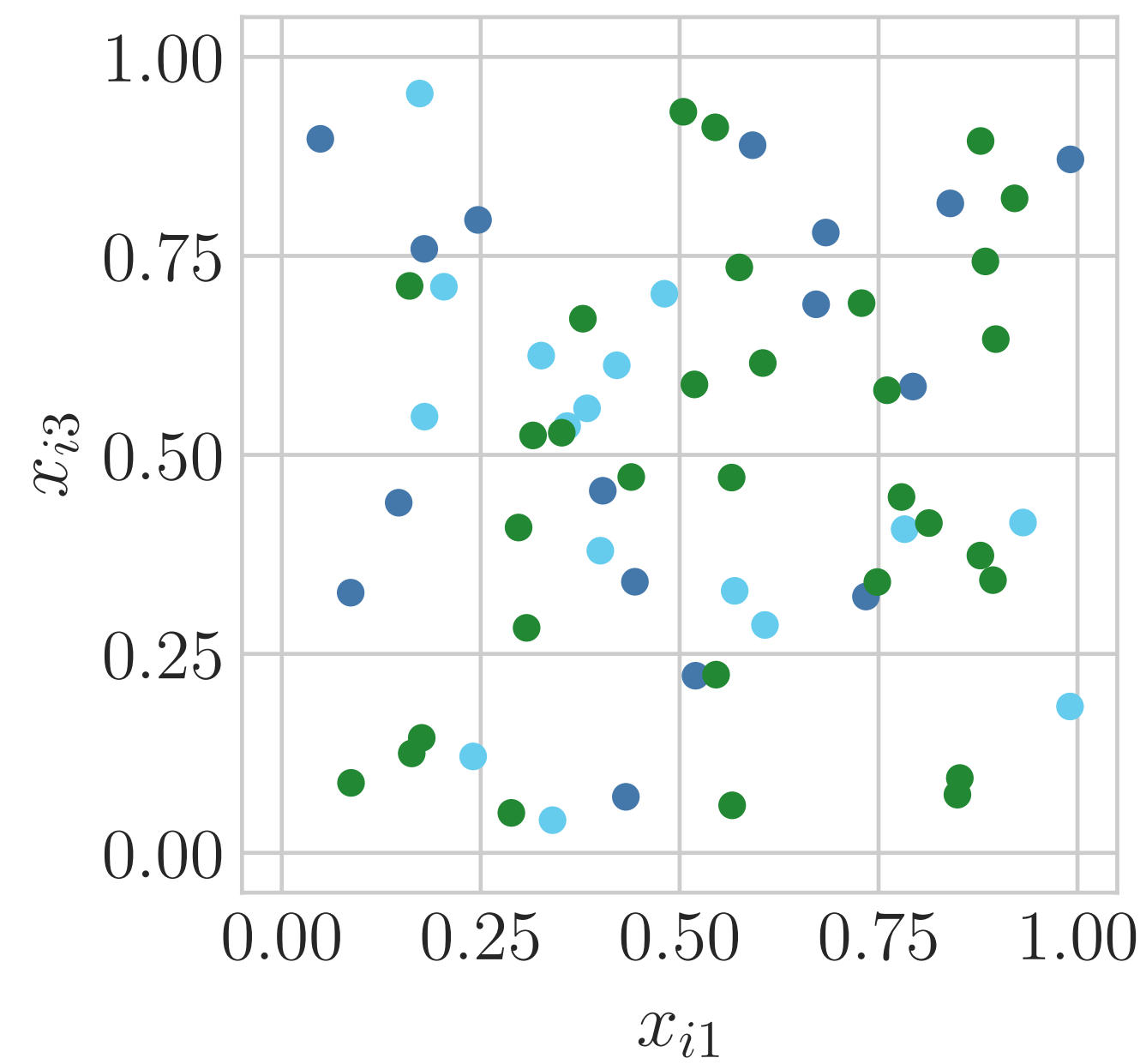
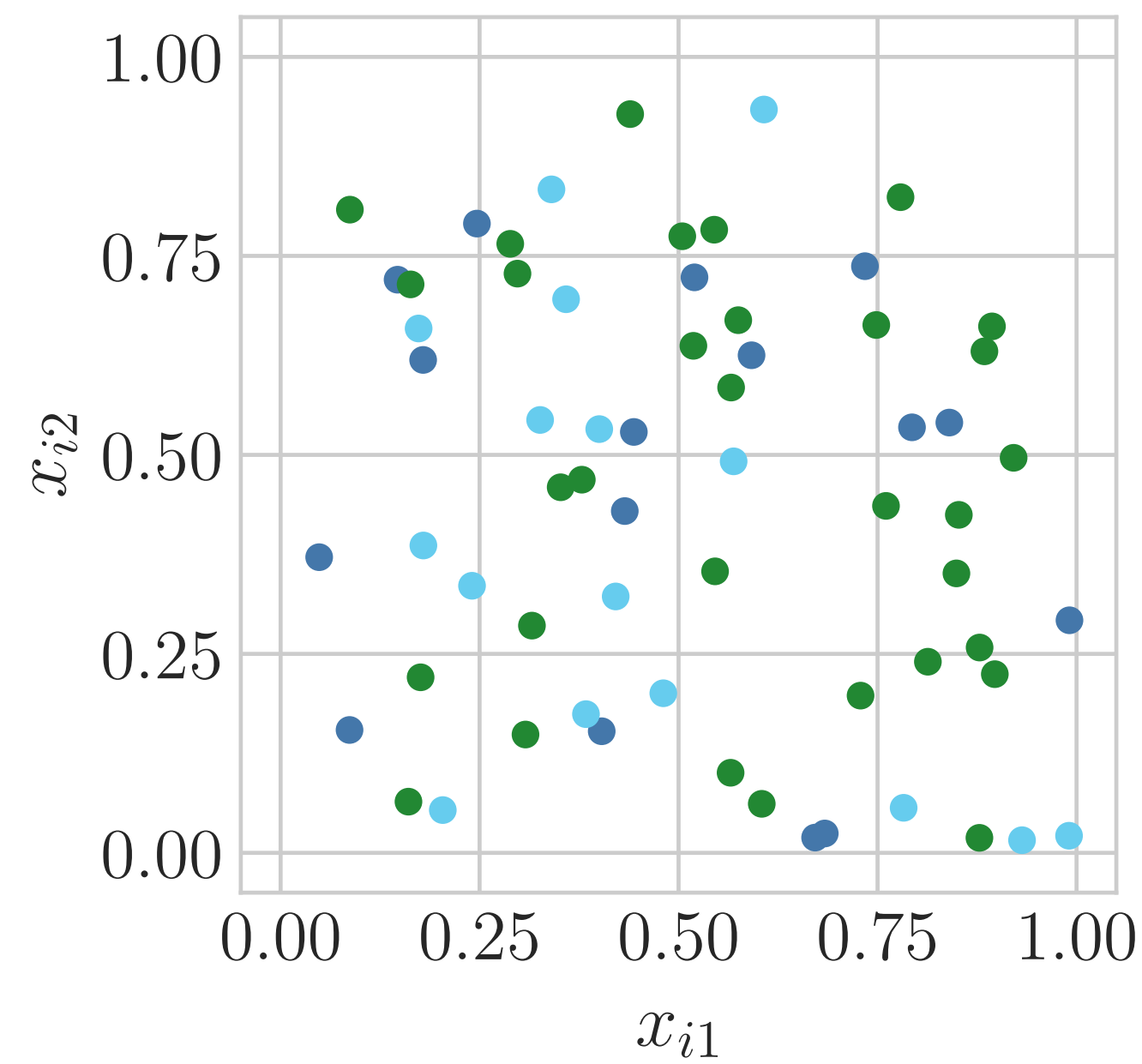


Four families of low discrepancy sequences

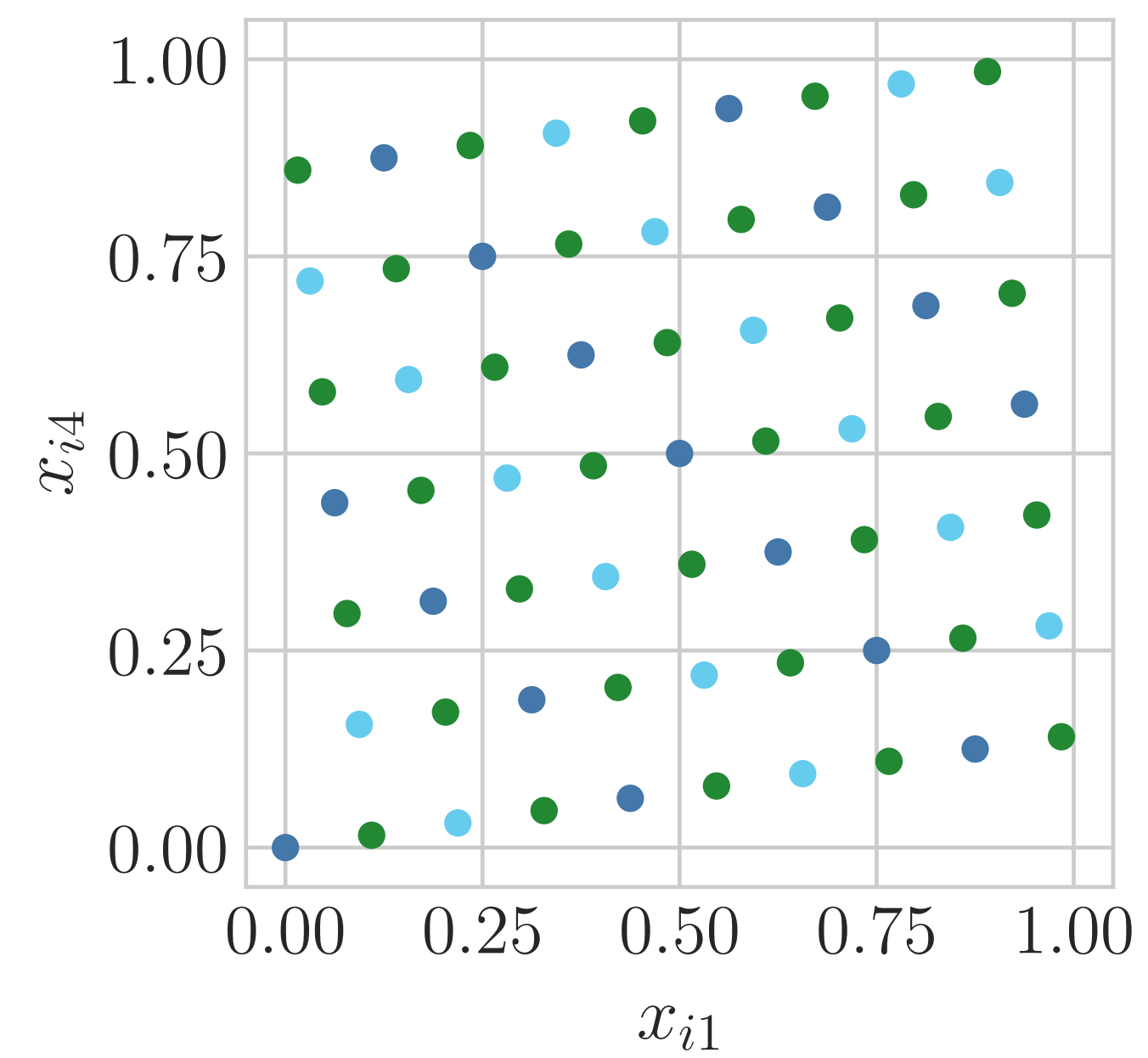
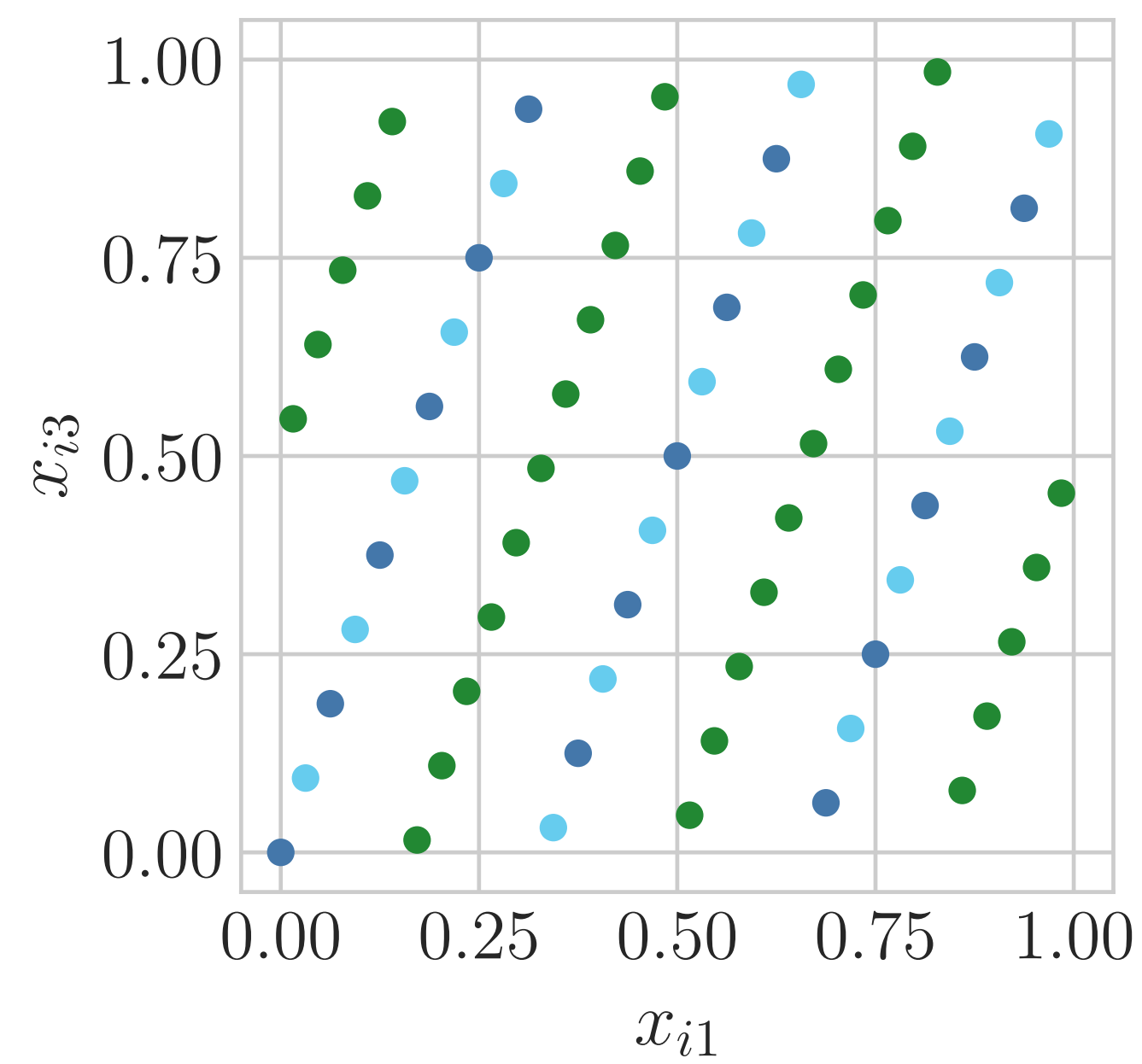
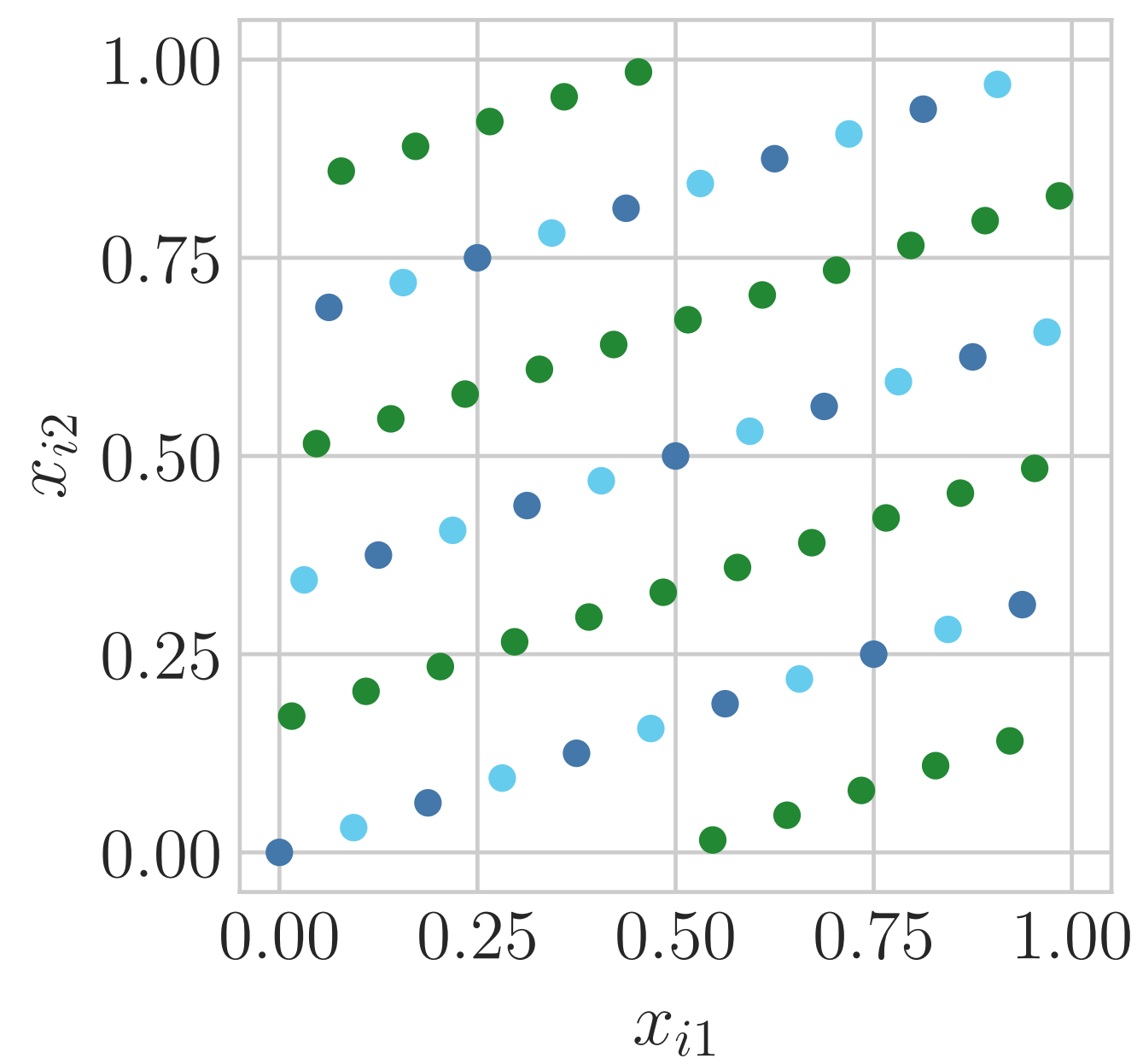
For $i = 0, 1, \dots$

- **Kronecker**: $x_i = i\zeta \bmod 1$ for some specially chosen generator $\zeta \in [0, 1)^d$,
no preferred sample size, n
- **Lattice**: $x_i = \phi_b(i)\zeta \bmod 1$ for some specially chosen generator $\zeta \in \mathbb{N}^d$,
preferred sample size is $n = b^m$
- **Halton**: $x_i = (\phi_2(i), \phi_3(i), \phi_5(i), \dots)$, essentially no preferred sample size, n
- **Digital**: $x_i = i_0\mathbf{c}_0 \oplus i_1\mathbf{c}_1 \oplus i_2\mathbf{c}_2 \oplus \dots$, where $i = (\dots i_2 i_1 i_0)_b$, $\oplus$ means
digitwise addition modulo b , and $\mathbf{c}_0, \mathbf{c}_1, \mathbf{c}_2, \dots \in [0, 1)^d$, preferred sample size
is $n = b^m$

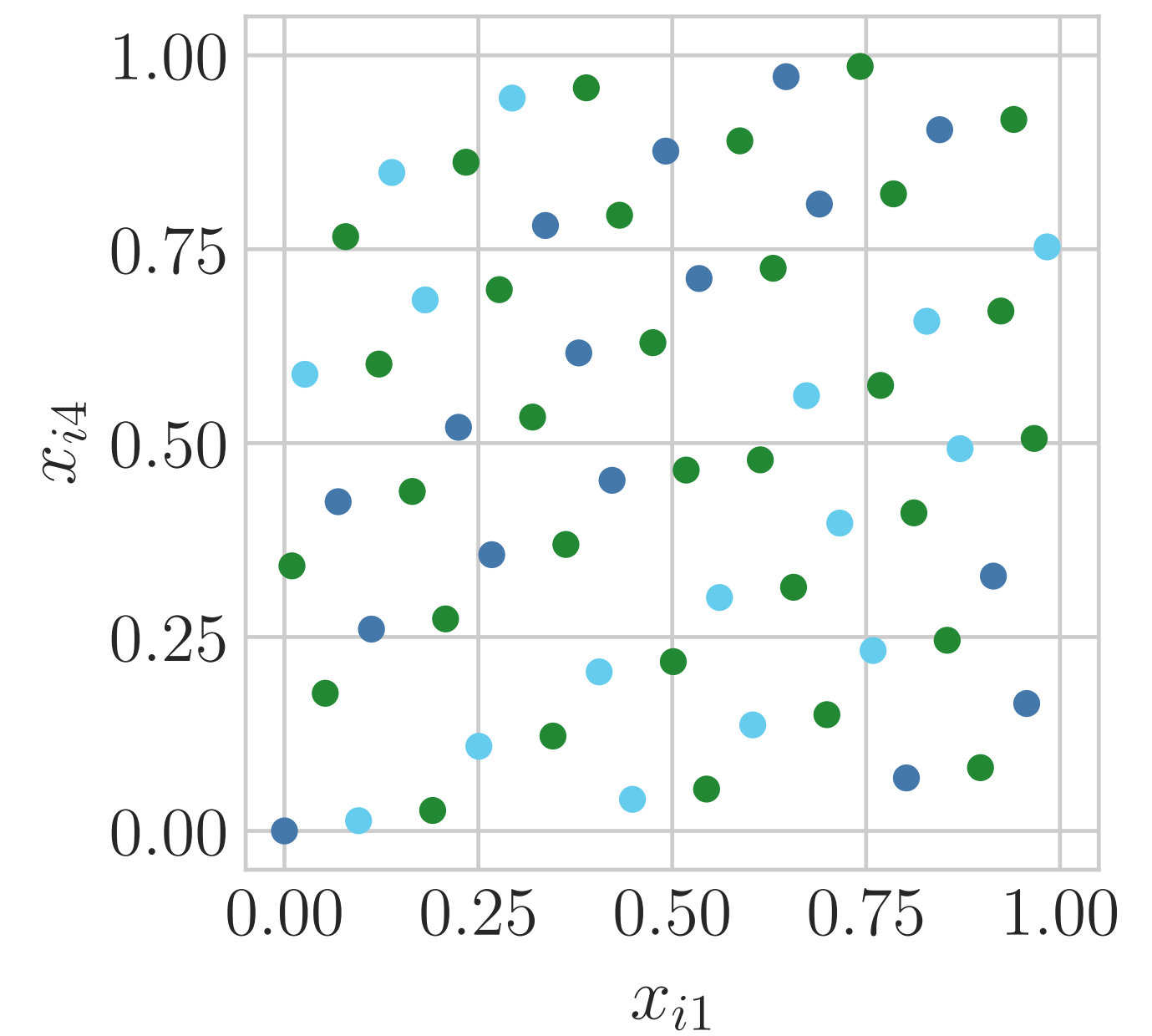
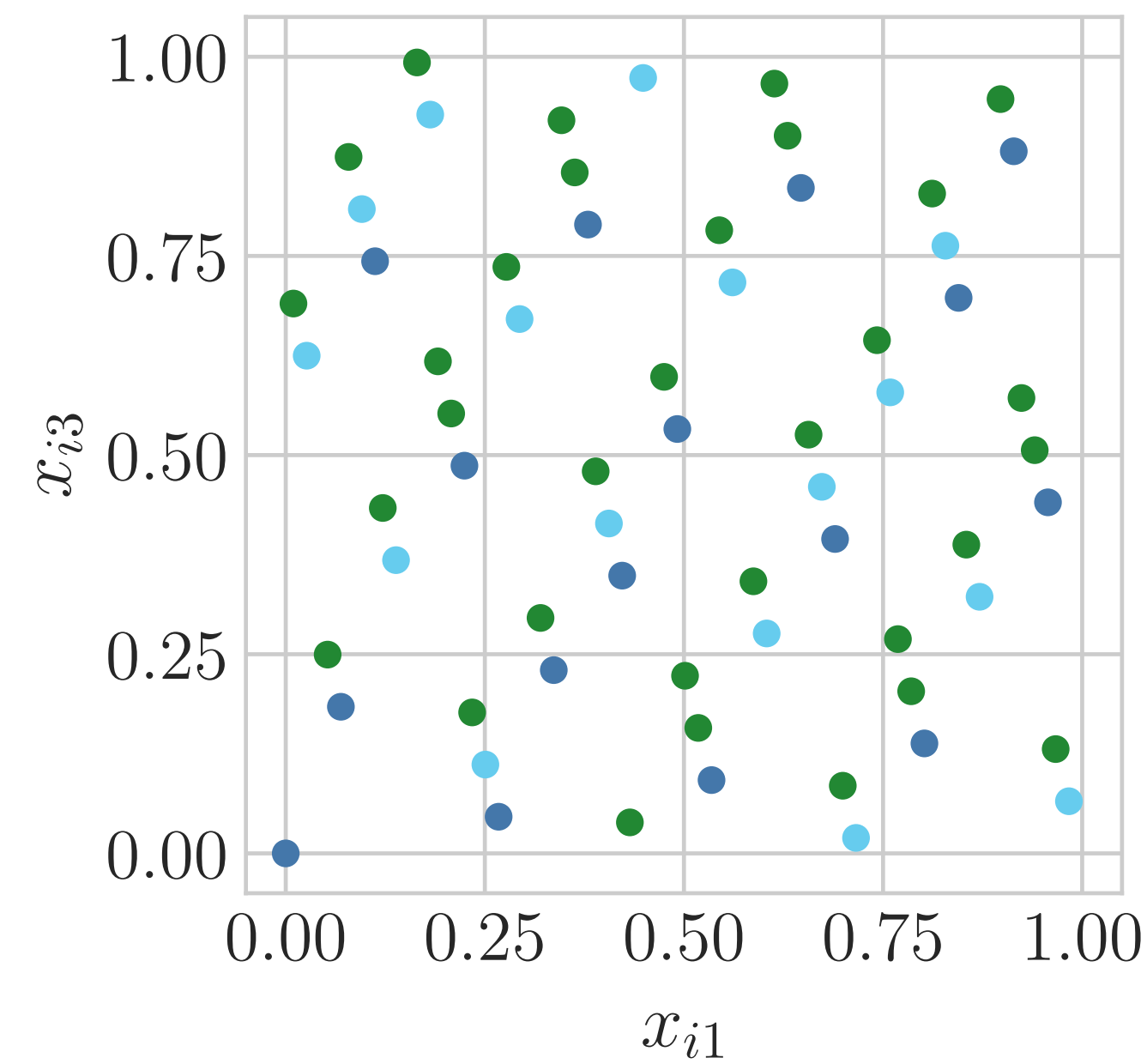
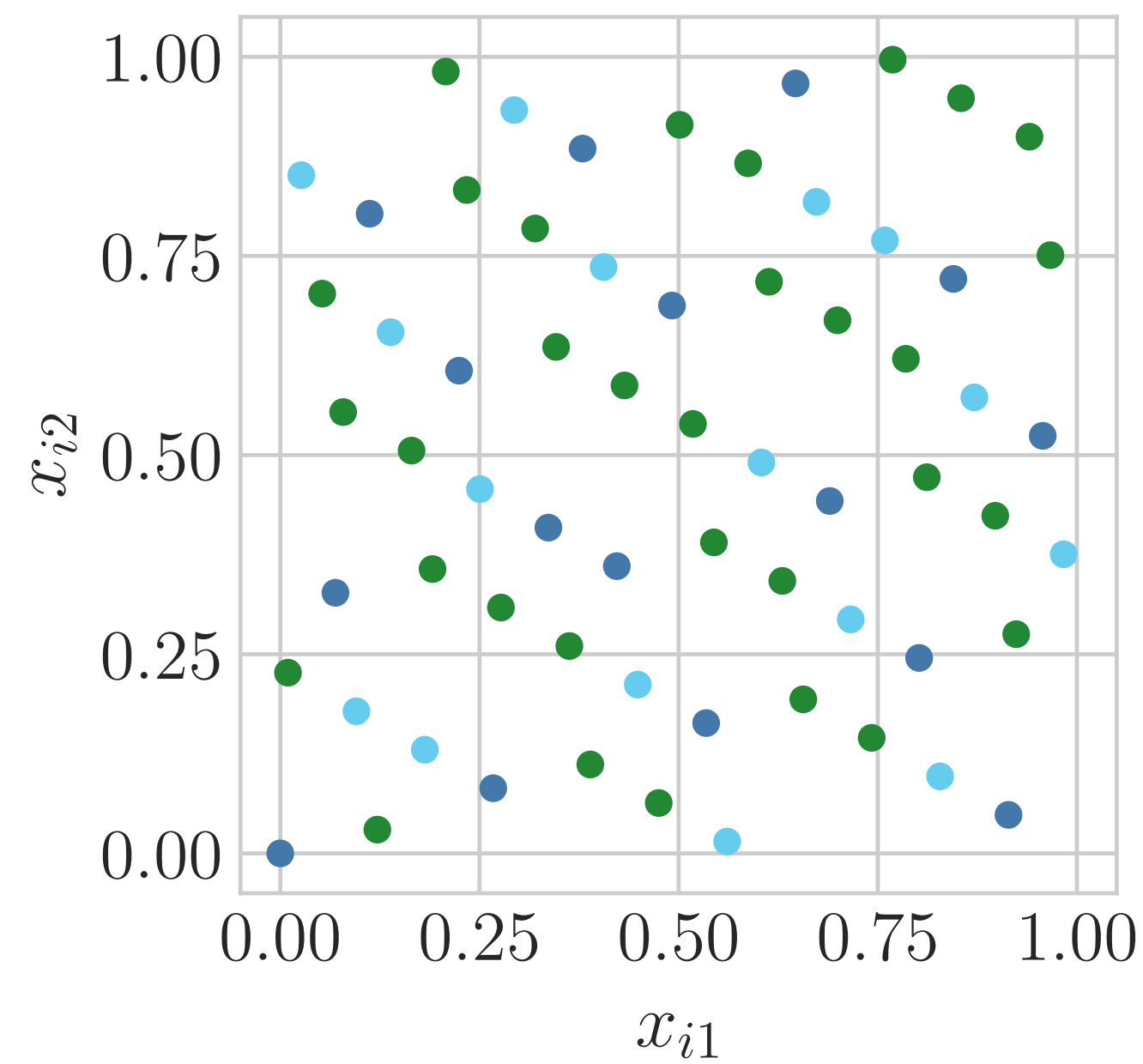
IID



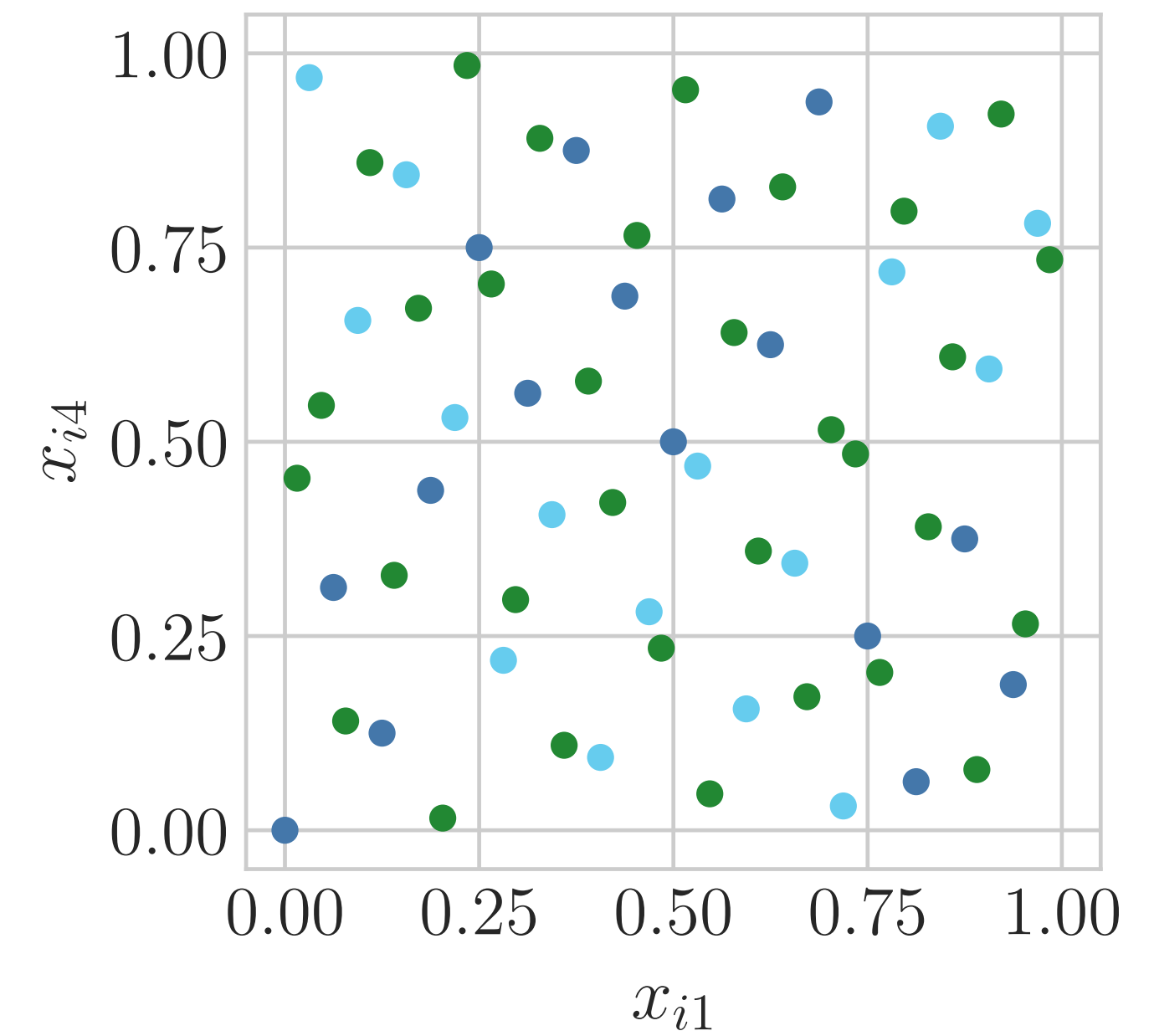
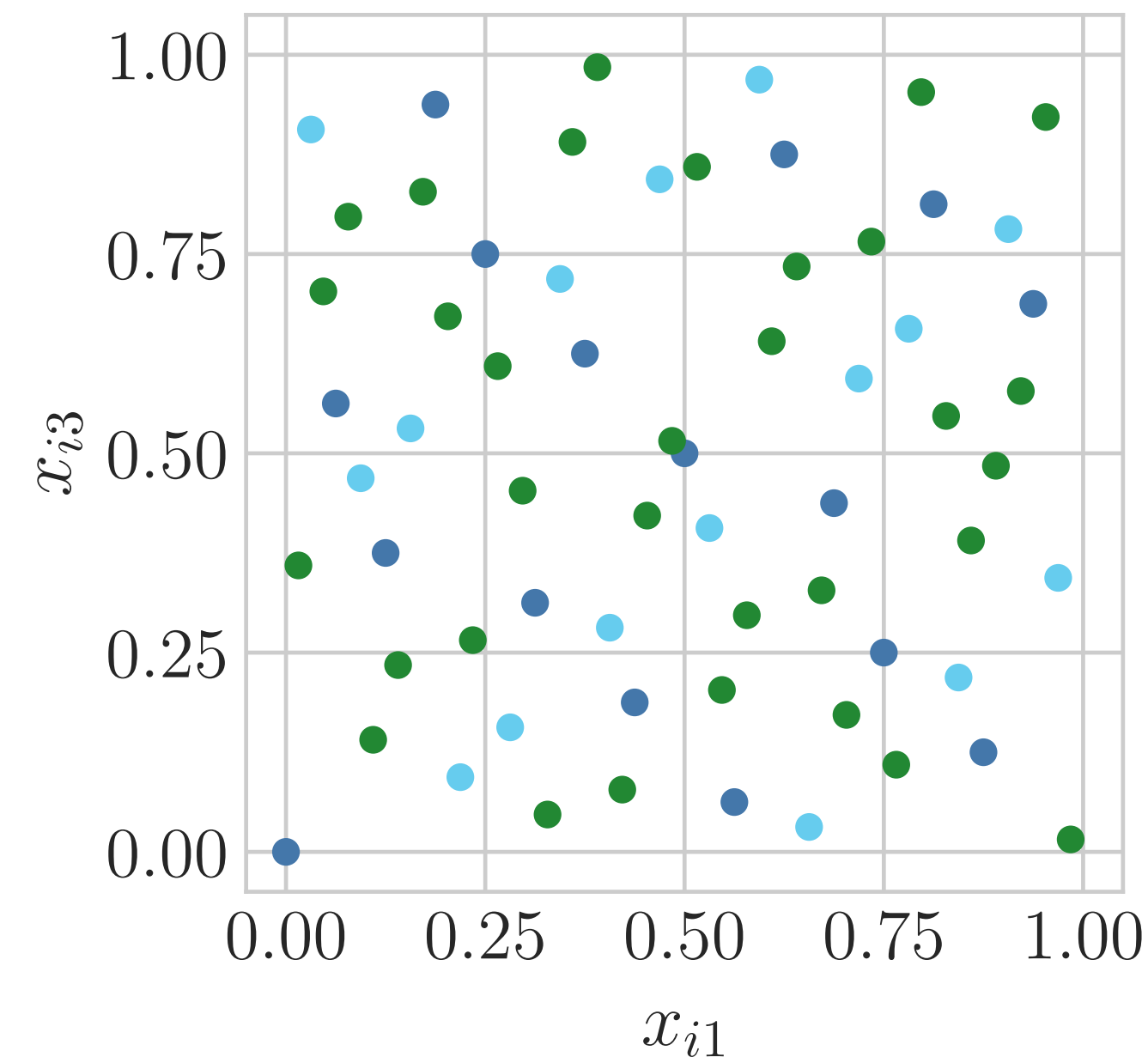
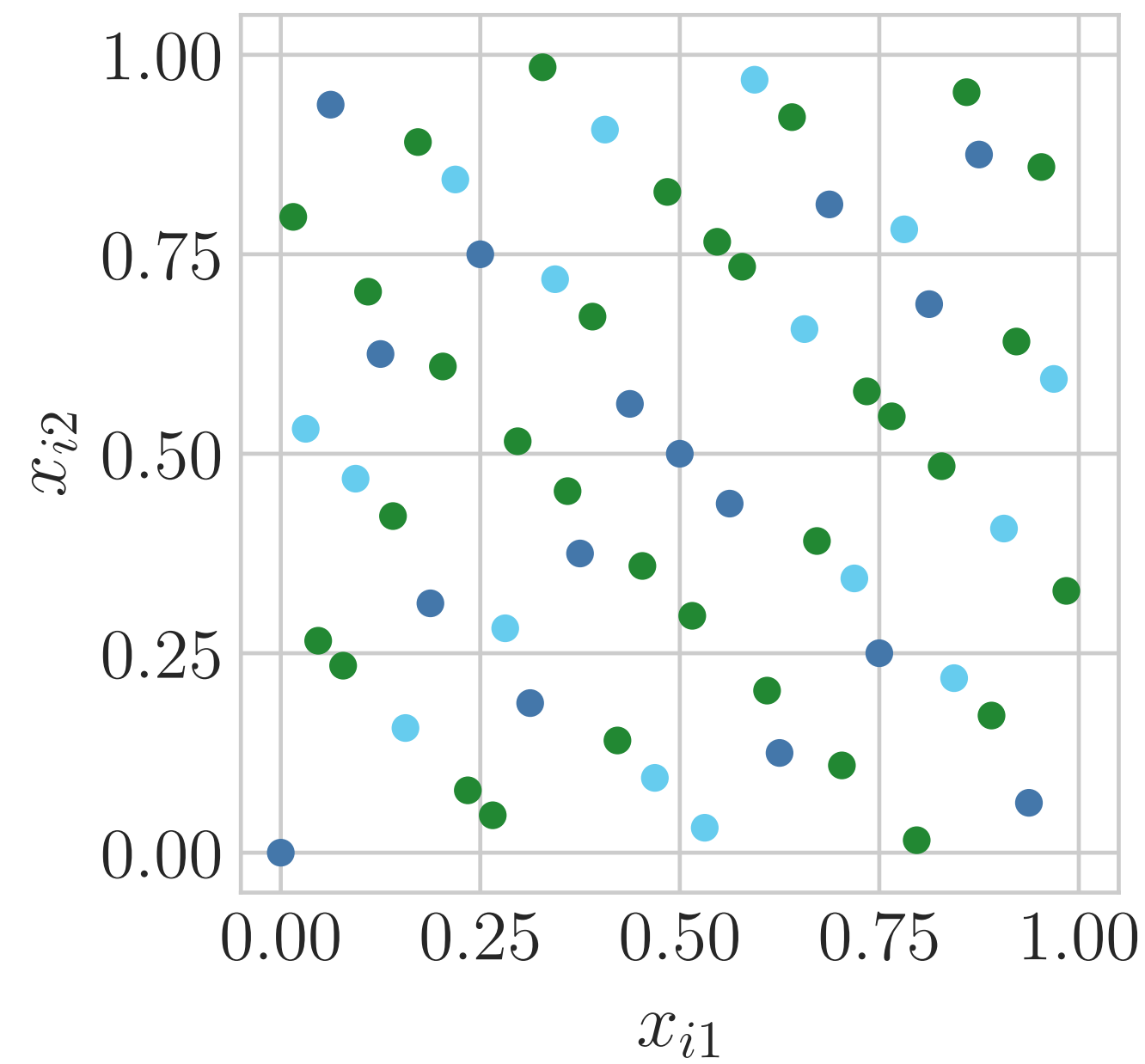
Lattice LD



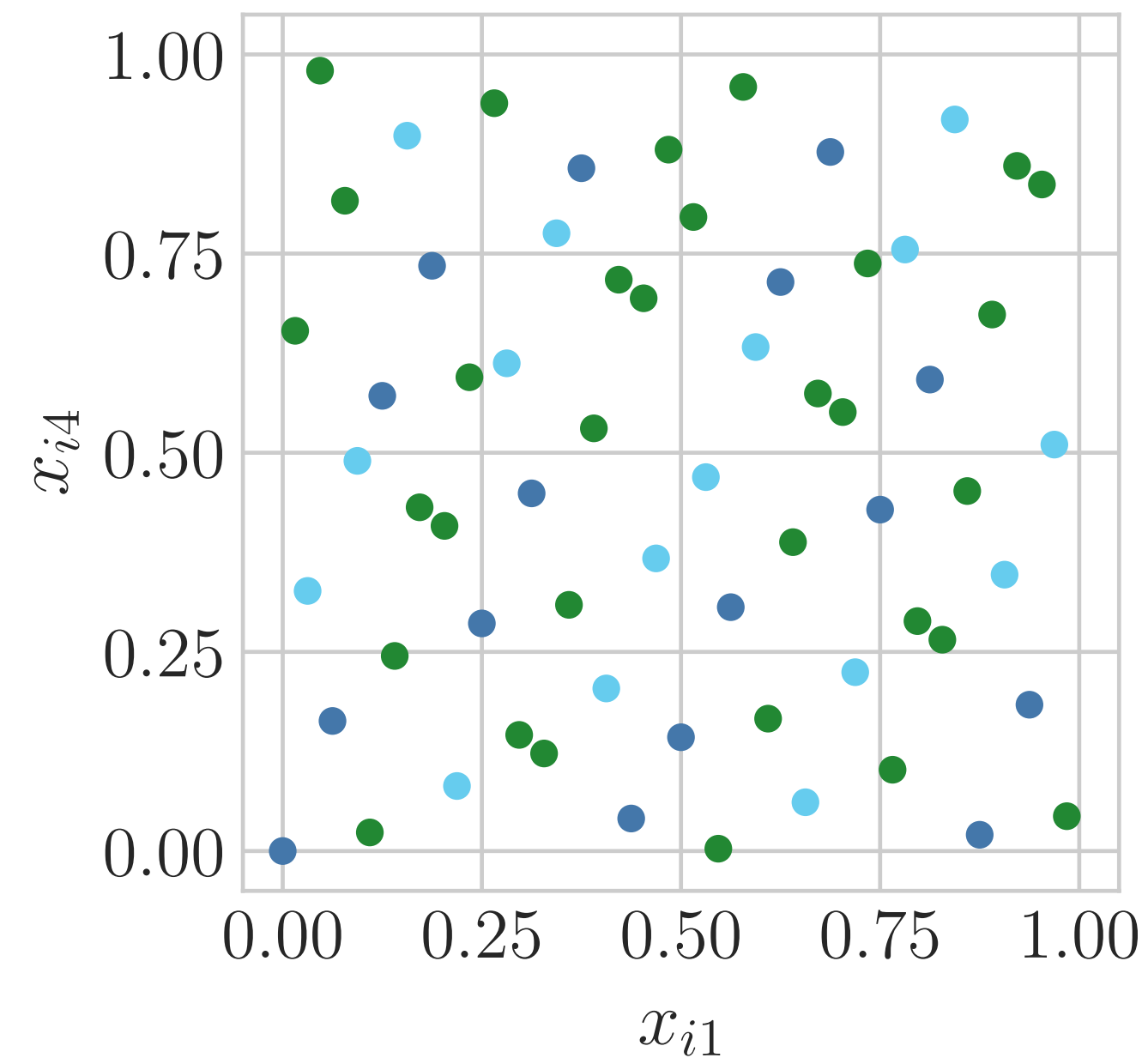
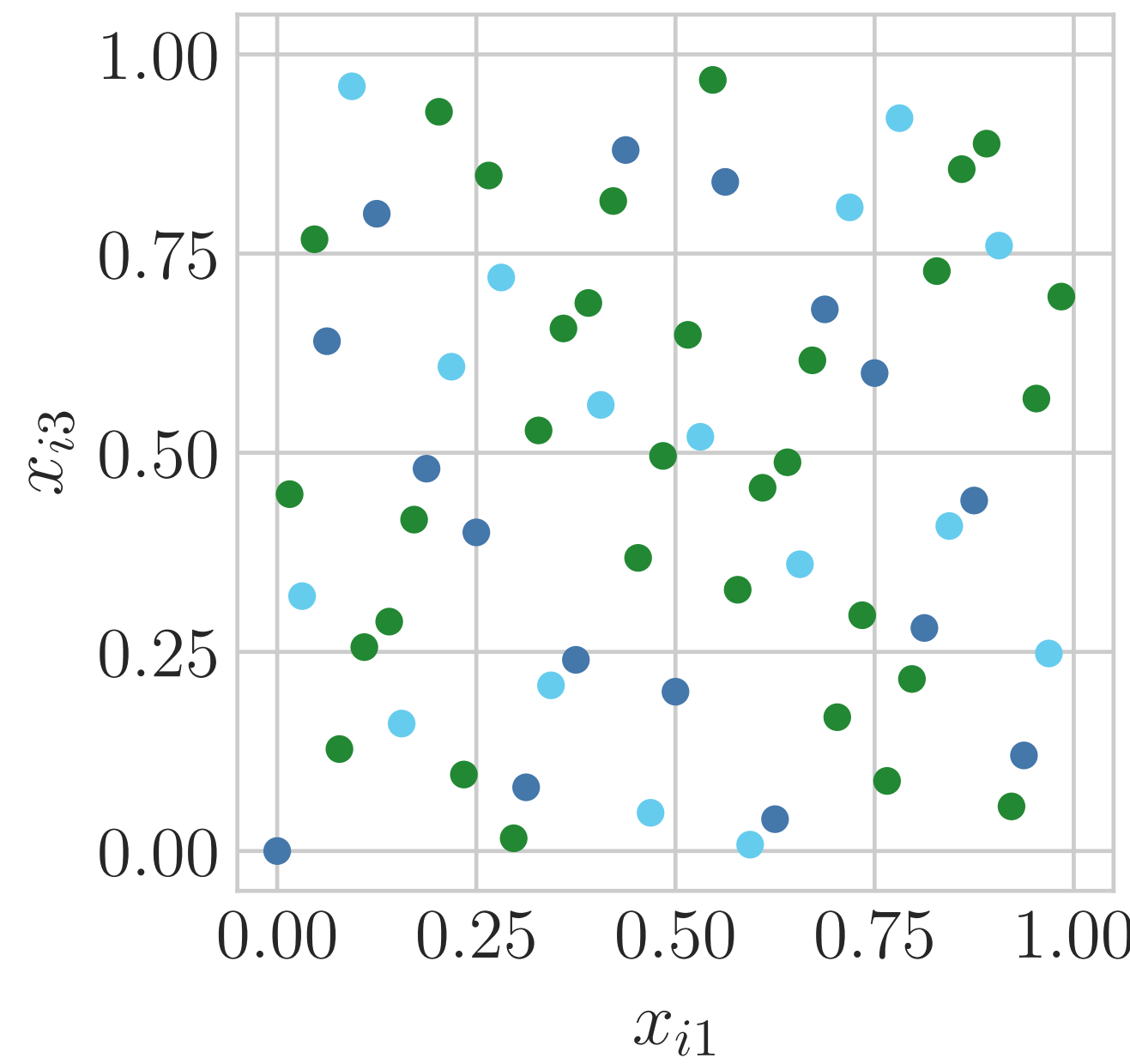
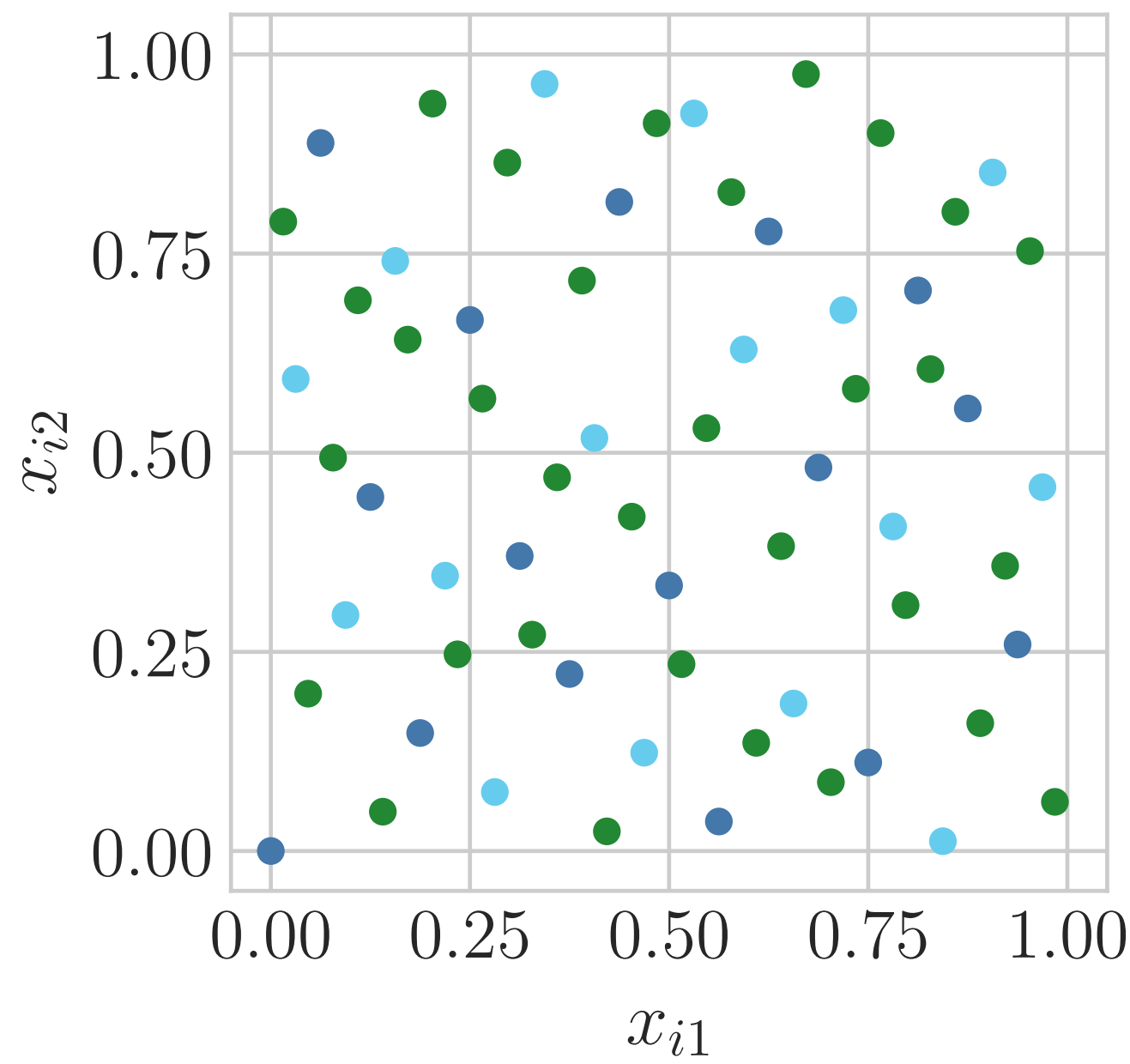
Kronecker LD

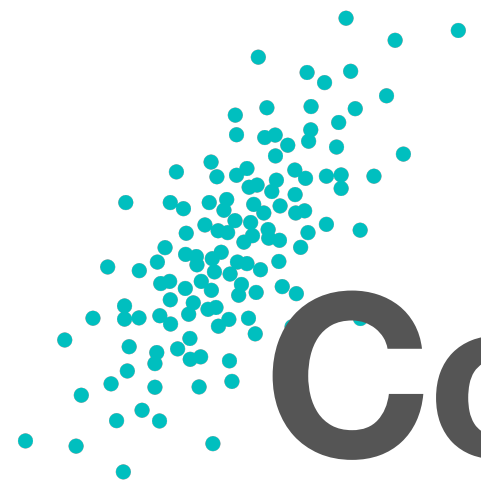


Sobol' LD



Halton LD





Component-by-component search

The generators for digital nets may be chosen by number theory, but in most cases the generators are chosen by computer search to minimize the discrepancy.

The component-by-component search method

1. Choose a generator for a 1-D sequence that minimizes the discrepancy over a range of n , set $\ell = 1$
2. Freeze the first ℓ dimensions of the generator
3. Choose the $\ell + 1$ component of the generator by minimizing the discrepancy
4. Increases ℓ by one
5. If $\ell < d$, go to step 2; otherwise you are done



Randomizing low discrepancy sequences

Want to make $\mathbf{x}_{\text{rand},i} \sim \mathcal{U}[0,1]^d$, while $\{\mathbf{x}_{\text{rand},i}\}_{i=0}^{n-1}$ is low discrepancy

- Random **shifts**: $\mathbf{x}_{\text{rand},i} = \mathbf{x}_i + \Delta \bmod \mathbf{1}$, $\Delta \sim \mathcal{U}[0,1]^d$, especially for lattices and Kronecker
- Random **digital shifts**: $\mathbf{x}_{\text{rand},i} = \mathbf{x}_i \oplus \Delta \bmod \mathbf{1}$, $\Delta \sim \mathcal{U}[0,1]^d$, especially for digital sequences
- **Linear matrix scrambling** or **nested uniform scrambling**, especially for digital sequences



Benefits of randomizing

- Unbiased estimator, $\hat{\mu} = \frac{1}{n} \sum_{i=0}^{n-1} f(\mathbf{x}_{\text{rand},i})$, of $\mathbb{E}[f(X)]$ for $X \sim \mathcal{U}[0,1]^d$
- $\mathbf{x}_{\text{rand},i}$ almost surely never on the boundary of $\mathcal{U}[0,1]^d$, so transformations to the Gaussian distribution will be finite
- Root mean squared error for digital sequences with linear matrix or nested uniform scrambling can be nearly $\mathcal{O}(n^{-3/2})$



Stopping Criteria

There is often the question of how big is your error for a given n ,

$$|\mu - \hat{\mu}_n| \leq ?$$

Or how big must n be so that the error is no greater than a tolerance, ε ,

$$|\mu - \hat{\mu}_n| \leq \varepsilon \text{ for } n = ?$$

We want to answer these questions based only on $f(\mathbf{x}_0), f(\mathbf{x}_1), \dots$

For IID sampling we might use the Central Limit Theorem (CLT). For LD sampling we can use the CLT only if we use replications. Other stopping criterion for LD sampling are based on

- Tracking the Fourier coefficients of the data, $f(\mathbf{x}_0), f(\mathbf{x}_1), \dots$, or
- Bayesian credible intervals